ElderCareAI User Guide

Follow the screens from sign in through caregiver monitoring and administrator functions. Each figure space names the screenshot to insert. Use the intended patient account and conceal private details. The Installation Guide appears at the end.

1 Sign in

Screen: Phone app opening screen

What to screenshot: Capture the Google sign-in button and app name.

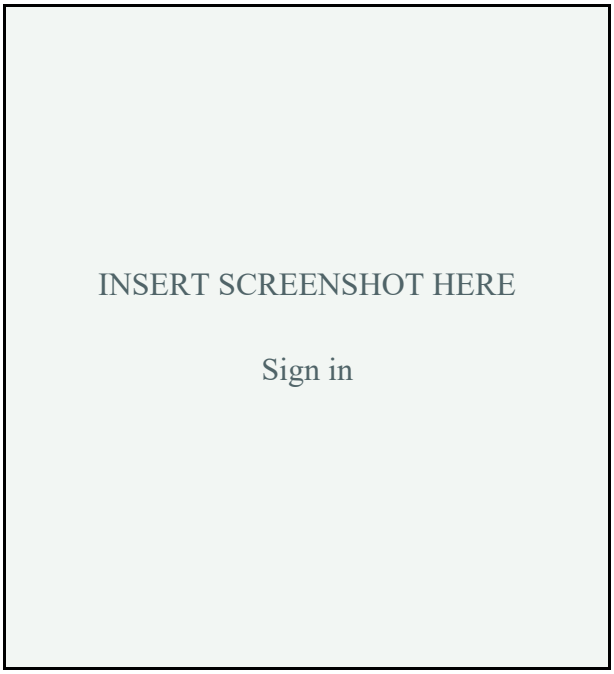


Figure 1: Sign in

1. Open ElderCareAI and tap Sign in with Google. Choose the caregiver account responsible for this patient.
2. Complete Google's account selection. If cancelled or unsuccessful, return to sign in and retry with internet available.
3. On first use, follow caregiver and patient setup. A returning account restores its saved setup and patient context.
4. Check the patient name on Home before viewing or recording care. Sign out through Profile before using another account.
5. Conceal the account email in screenshots. A saved session may bypass sign in on subsequent launches.

2 Caregiver and patient setup

Screen: First sign-in onboarding

What to screenshot: Capture the caregiver profile and patient profile screens separately.

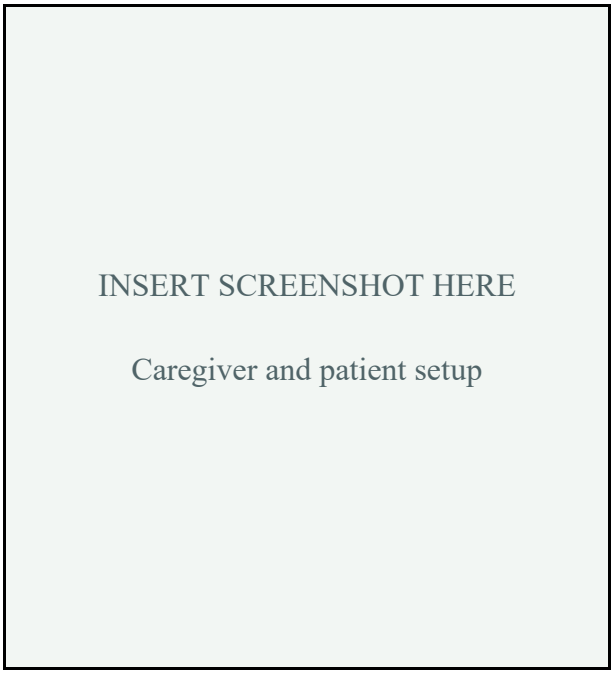


Figure 2: Caregiver and patient setup

1. Enter the caregiver name, contact number, age and gender; the signed-in email identifies the account. Add a photo if desired.
2. Continue to the patient form and enter the older adult's name, birth date or age, gender, and the requested health details.
3. Add weight, height, blood type, known conditions and current medications when available. Use the displayed units.
4. Enter the emergency contact details and save. Correct any highlighted invalid field before continuing.
5. Patient selection controls the vitals, sleep, care plan, location and alerts displayed. Confirm the correct patient after changing accounts.

3 Connect Galaxy Watch6

Screen: Profile or setup watch integration

What to screenshot: Capture Connect or change watch and the paired device status.

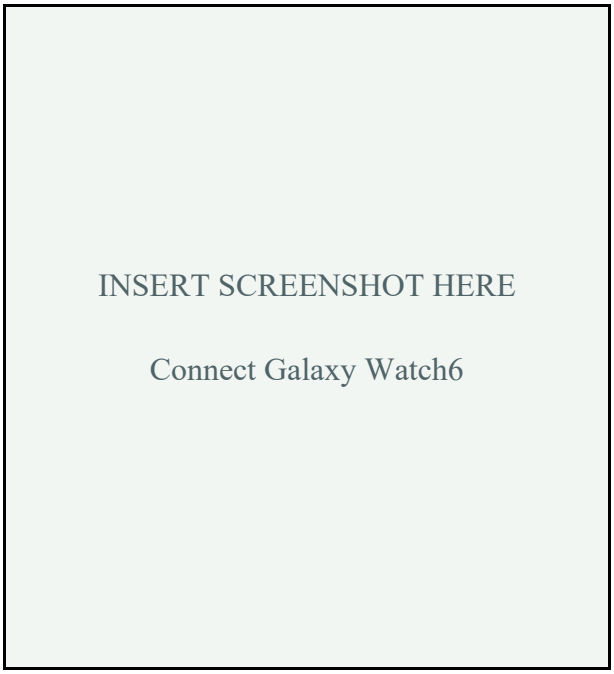


Figure 3: Connect Galaxy Watch6

1. Open Profile and choose Connect or change watch, or continue from watch setup during onboarding.
2. Generate the eight-digit pairing code for the intended patient. Open ElderCareAI on the watch and enter that code.
3. Tap Pair watch and wait for confirmation. An invalid or expired code requires a new code; network errors require a working connection.
4. Confirm the phone shows Galaxy Watch6 connected. This app pairing is separate from Galaxy Wearable's Bluetooth pairing.
5. Allow watch sensor, activity, location and notification permissions, then start monitoring. Pair the watch again when intentionally assigning it to a different patient.

4 Watch monitoring

Screen: ElderCareAI on the watch

What to screenshot: Capture Start monitoring or Stop monitoring, current readings, Location and sharing status.

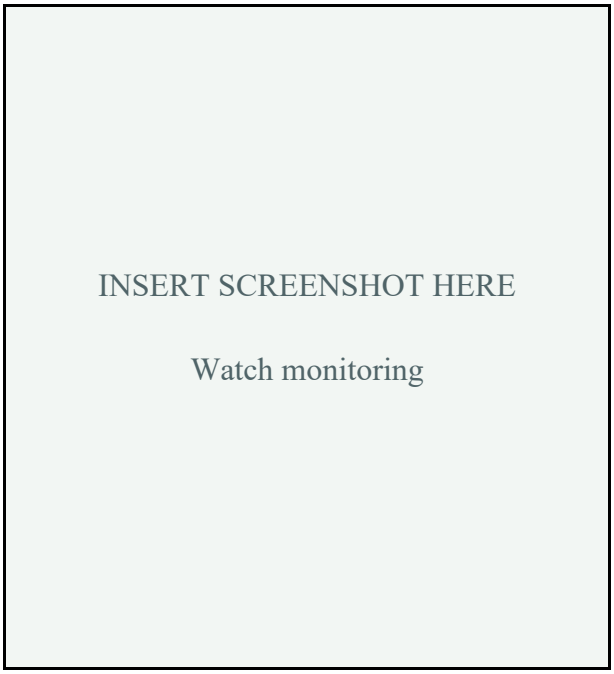


Figure 4: Watch monitoring

1. Open ElderCareAI on the paired Galaxy Watch6 and tap Start monitoring. Allow requested permissions and keep the watch snug on the wrist.
2. Heart rate, skin temperature, steps and HRV update when usable sensor events arrive. HRV needs a sustained interval of valid beat data and may restart after poor contact.
3. Location starts with monitoring. Enable Location and internet; move outdoors for better satellite reception. GPS, fused and network estimates retain source and accuracy labels.
4. Open Measure blood oxygen for a manual spot check. Keep the arm still. Optional five-minute auto checks work only while that screen stays open.
5. Inspect sharing status and queued samples. Offline vital measurements retry when connected. Tap Stop monitoring to end monitoring; restart it after a watch reboot or force stop.

5 Caregiver dashboard

Screen: Home tab

What to screenshot: Capture live vital cards with visible values, labels and measurement times.

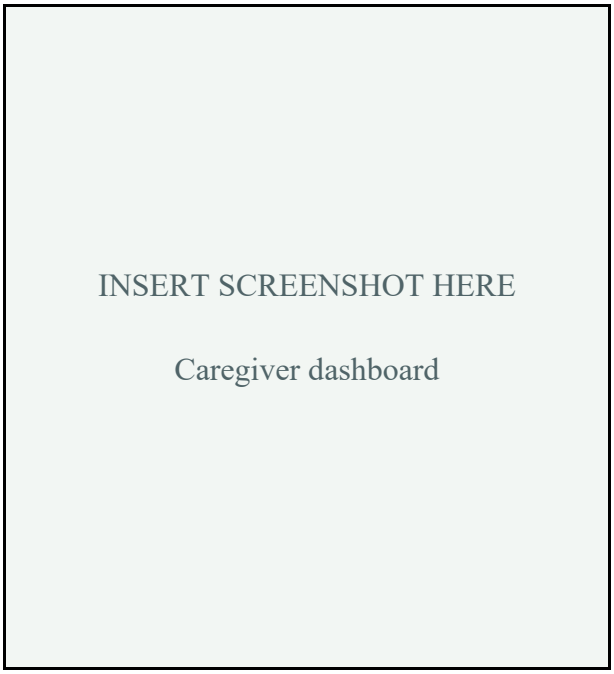


Figure 5: Caregiver dashboard

1. The dashboard summarizes the selected patient. Each vital displays its latest available reading and a text status.
2. A fresh heart-rate update can change the display label without reopening the screen. The alert rule remains above 110 BPM.
3. A waiting or stale label means a fresh measurement is unavailable; it is not a normal reading.
4. Tap Sync now or pull to refresh when needed. A failed refresh should not be mistaken for a new measurement.
5. Tap a vital card for detail or a care task to open its relevant Care plan screen.

6 Vital signs

Screen: Health tab Vitals

What to screenshot: Capture heart rate, oxygen, sleep, steps, skin temperature and stress if present.

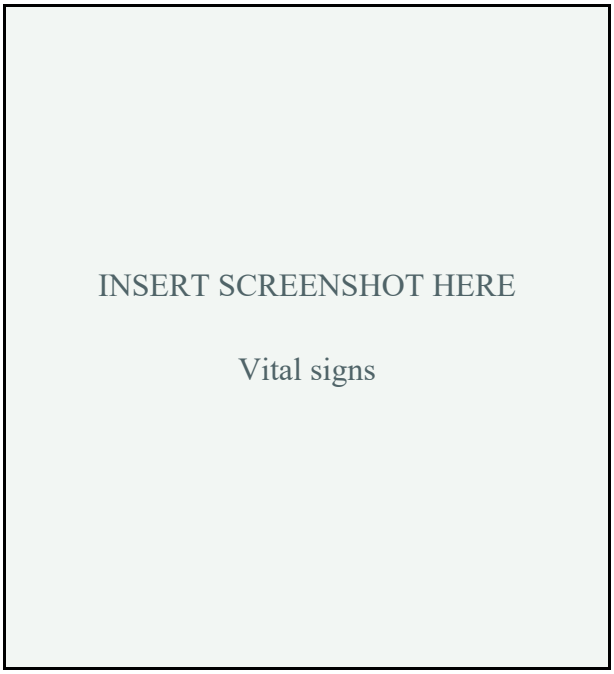


Figure 6: Vital signs

1. Open Health and select Vitals. Review Heart Rate, Blood Oxygen, Skin Temperature, Stress level, Sleep and Steps.
2. Each available card shows a value, unit, status and measurement time. A status can change with a new reading; missing or stale values are not normal readings.
3. Heart rate, wrist temperature and valid HRV come from watch monitoring. Steps count the increments collected during monitoring on that watch day.
4. Blood oxygen uses spot measurements or available imported history. Sleep appears after Samsung Health records and synchronizes the patient's session.
5. Stress is an experimental HRV-based index. It needs a personal reference before a score is available and is not a medical diagnosis.

7 Vital detail and history

Screen: Tap a vital card

What to screenshot: Capture one vital detail with its dated reading and status explanation.

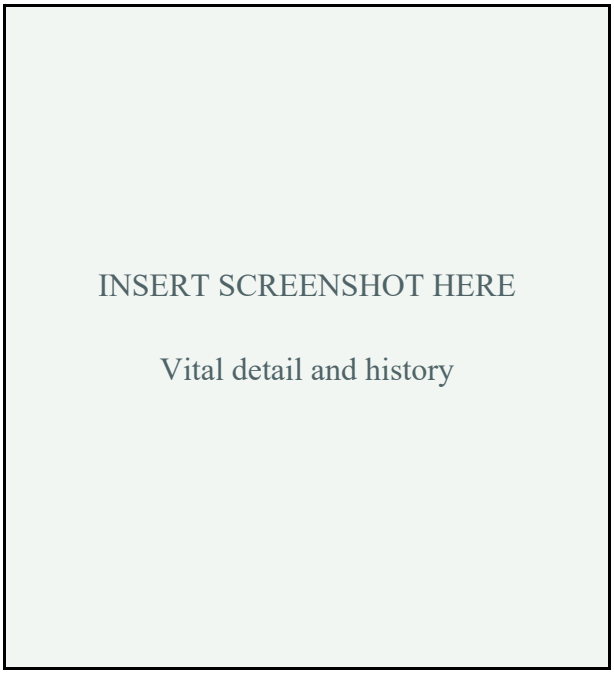


Figure 7: Vital detail and history

1. Tap any vital card to open its details. Read the latest value, status and original measurement time.
2. Scroll to Elle insight. Its observation remains steady while the live value above continues updating.
3. When New readings available appears, tap Refresh insight when ready. The Reading reviewed timestamp identifies the observation's reference reading.
4. Review recent measured history or nightly sleep entries, source details and the explanation below the card.
5. If the insight request fails, use Retry insight. Interpret status labels and wellness suggestions as review guidance; confirm concerns with an appropriate professional.

8 Health analytics

Screen: Health tab Analytics

What to screenshot: Capture the date filter, line chart, bar chart and a selected timestamp.

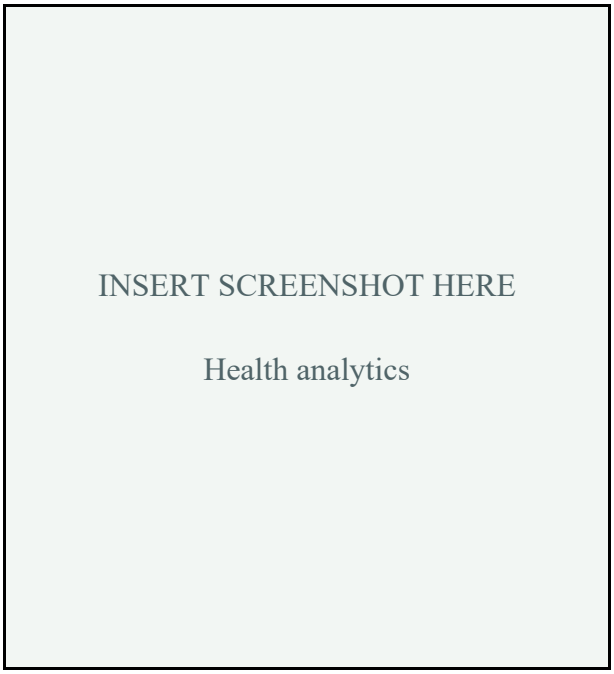


Figure 8: Health analytics

1. Choose a metric and date range. The chart uses saved database readings for the selected patient.
2. Switch between line and thin bar views. Inspect a point to see its original value and measurement time.
3. Daily steps show a day total rather than adding cumulative samples together.
4. Use actual measurement timestamps to distinguish delayed uploads from newly measured values.
5. If the range is empty, choose a date with saved readings. Retry a failed load instead of assuming a zero value.

9 Health history sync

Screen: Profile or Settings Health History

What to screenshot: Capture the sync status and one saved sleep history entry.

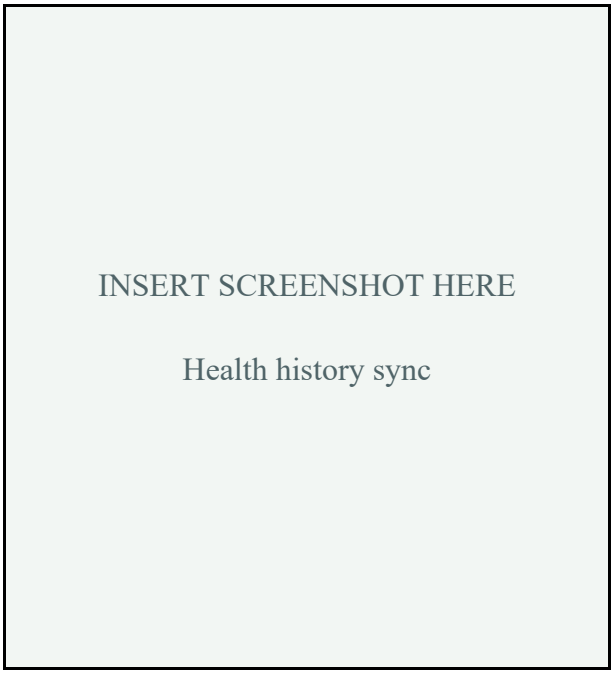


Figure 9: Health history sync

1. On the same Android phone paired to the patient's watch, install and open Samsung Health and allow its watch synchronization.
2. In ElderCareAI Profile, open Health History and connect the history import. Grant the requested access when prompted.
3. Tap Sync health history after a recorded sleep session is available. Importing sleep is different from live watch sensor monitoring.
4. Check the sync result and Sleep card history. No new data means no additional source records; previously saved patient history should remain available.
5. If access or upload fails, check internet and Samsung Health permissions and retry. Developer installations may need Samsung Health Data SDK developer access. Background checks depend on Android scheduling; wake-up delivery is not immediate.

10 Live location and safe zone

Screen: Live location tracking

What to screenshot: Capture the watch marker, accuracy or source label, and safe-zone circle.



Figure 10: Live location and safe zone

1. On the watch, tap Start monitoring with Location and internet enabled. Open Live location tracking on the phone; no separate walking mode is needed.
2. Read the marker's source, measurement age and accuracy. The displayed stream can contain watch GPS, fused and network estimates; fused location may represent the paired phone.
3. Open Set up safe zone or Edit safe zone. Enter a place name, tap Choose on map and tap the actual home center, including inside the circle or on the marker.
4. Choose a radius from 10 to 5,000 meters, then Save and enable safe zone. Use current location can copy a fresh usable estimate; the saved center stays fixed.
5. An estimated exit uses 3 qualifying outside updates spanning at least 20 seconds. For a 10 or 20 m zone, reported accuracy must be 50 m or better. Larger-zone limits rise to at most 100 m.
6. GPS, fused and network can all confirm the marker-based crossing. Very inaccurate or stale estimates cannot. Read confirmation progress and the source-labelled alert; small-zone false alarms are possible.
7. Keep the zone enabled and allow background notifications. Walk outside within internet coverage and inspect Alerts. Pan to explore or tap Follow location to return; Pause alerts disables boundary checks.

11 Medication list

Screen: Care plan Medications

What to screenshot: Capture one medication card showing image, 12-hour time and Taken status.

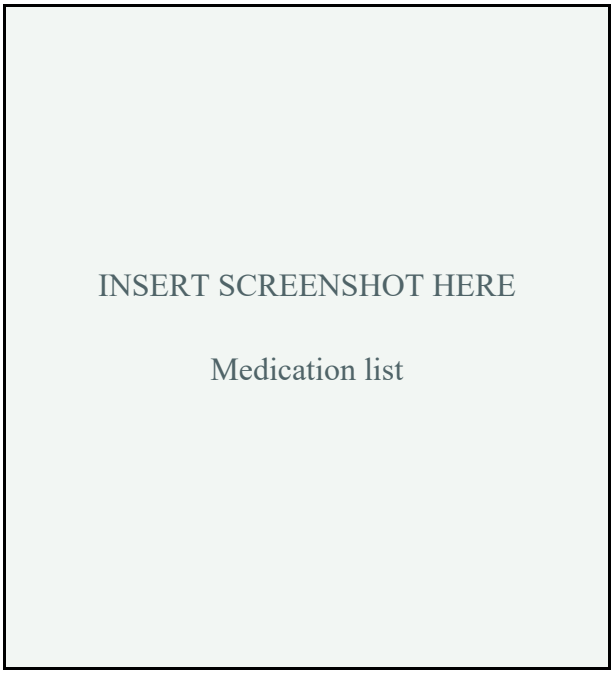


Figure 11: Medication list

1. Open Care plan then Medications. Tap Add to enter name, dosage, frequency and instructions from the verified prescription.
2. Choose an optional medication photo and review its preview. Without a photo the card uses its placeholder.
3. Select start and end dates and each reminder time. Displayed times use AM or PM; verify the device's local date and timezone before saving.
4. Save and check the medication card. Use Edit to correct the schedule; confirm Delete to remove it and cancel applicable reminders.
5. Tap Taken only after the dose was taken, and select the corresponding dose when asked. Watch confirmations also update the phone after successful sync; pending offline confirmations are not yet saved.

12 Prescription draft with Elle

Screen: Care plan Scan or upload prescription

What to screenshot: Capture the Elle entry card and the editable extracted draft.

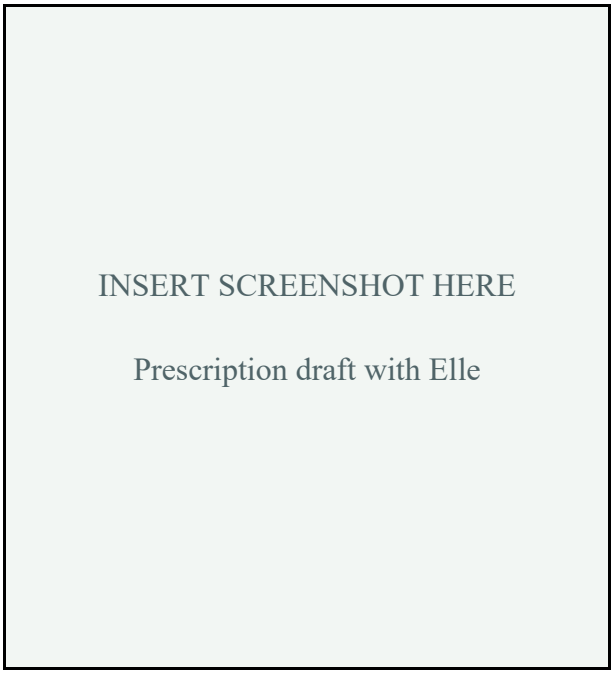


Figure 12: Prescription draft with Elle

1. Tap Scan or upload a prescription in Care plan. Choose a clear image or capture the original with the camera.
2. Wait for Elle to prepare an editable draft. If extraction fails or handwriting is unclear, retry with a clearer image or enter the medication manually.
3. Compare every extracted name, dosage, frequency, time, instruction, date and prescriber against the original. Correct mistakes and complete missing information.
4. Check the review confirmation only after verification. AI extraction does not prescribe medication or verify the doctor's instruction.
5. Confirm and save medication. Check the resulting medication card, schedule and reminder times; unconfirmed drafts do not activate a schedule.

13 Appointments

Screen: Care plan Appointments

What to screenshot: Capture a saved upcoming appointment showing patient, purpose, date and time.

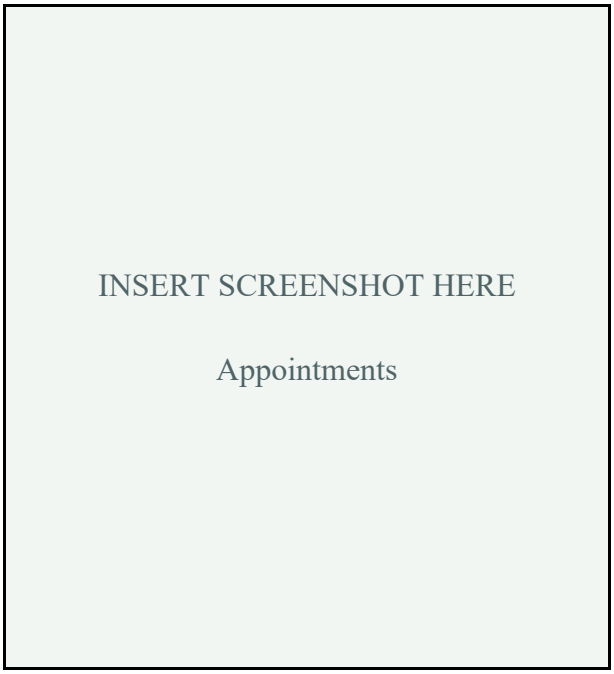


Figure 13: Appointments

1. Open Care plan then Appointments. Tap Add and enter the purpose or title, appointment date and local time.
2. Add doctor name, location and notes if known. Save, correcting any invalid field while keeping the draft available.
3. Review the saved visit. Appointment reminders are prepared ahead of the visit; the server schedule uses a 15-minute advance, so create a test visit at least 20 minutes ahead.
4. Enable background notifications and check phone time and internet. A reminder is not proof the appointment has started or been attended.
5. Use Edit to reschedule, Complete after attendance, or confirm Delete when cancelled. The list and combined schedule refresh after the change.

14 Care schedule

Screen: Care plan Schedule

What to screenshot: Capture two scheduled items in chronological order.

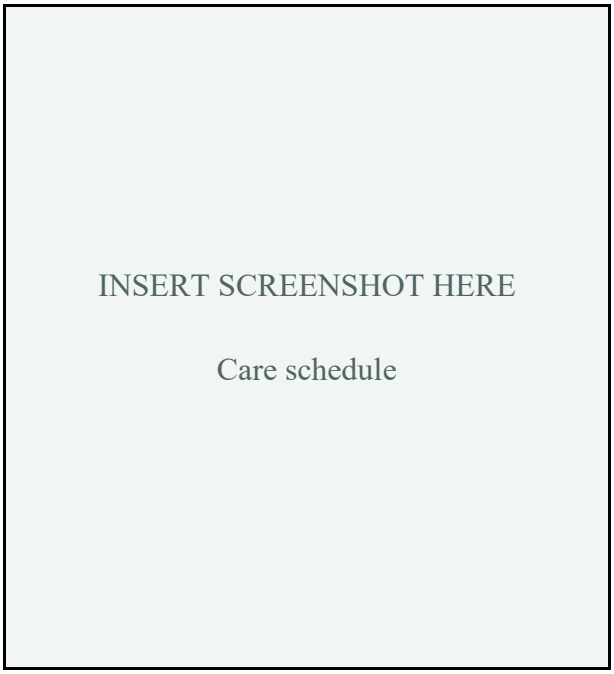


Figure 14: Care schedule

1. Open Schedule to review medications and appointments together. Items on the same day sort by their actual time.
2. Completed activities have a different status from upcoming activities.
3. When two reminders share one time, dismiss the first popup and capture the second popup separately.
4. Reschedule a visit or edit a dose time and reopen Schedule to check its new chronological position.
5. For simultaneous reminders, verify both separate items remain visible and capture both popups after dismissing each in turn.

15 Alerts and notifications

Screen: Alerts tab

What to screenshot: Capture collapsed Automatic vital alerts, SOS action, and one dated notification.

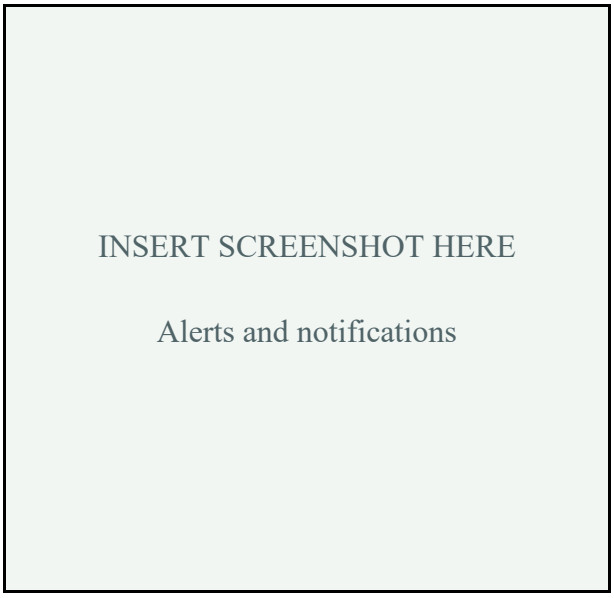


Figure 15: Alerts and notifications

1. Open Alerts and expand Automatic vital alerts to review the configured rules; collapse it to leave emergency actions visible.
2. A qualifying measured crossing saves a patient alert. The high-heart-rate warning starts above 110 BPM, rather than at 100 or exactly 110 BPM.
3. Open Notifications for medication, appointment and other notices. Review the timestamp and related details before acting.
4. Tap Enable background notifications and allow Android permission. Sound also needs the relevant channel enabled, audible volume and suitable Do Not Disturb settings.
5. When multiple alerts arrive together, dismiss or open the current popup to review the next queued popup. A linked-medication action is shown only when a relevant medication is associated.
6. Push can work when the app is backgrounded or normally closed; Android force-stop and device restrictions can prevent delivery until the app is opened again.

16 Emergency contacts and SOS

Screen: Alerts and Settings

What to screenshot: Capture contact order with numbers covered, plus the emergency action.

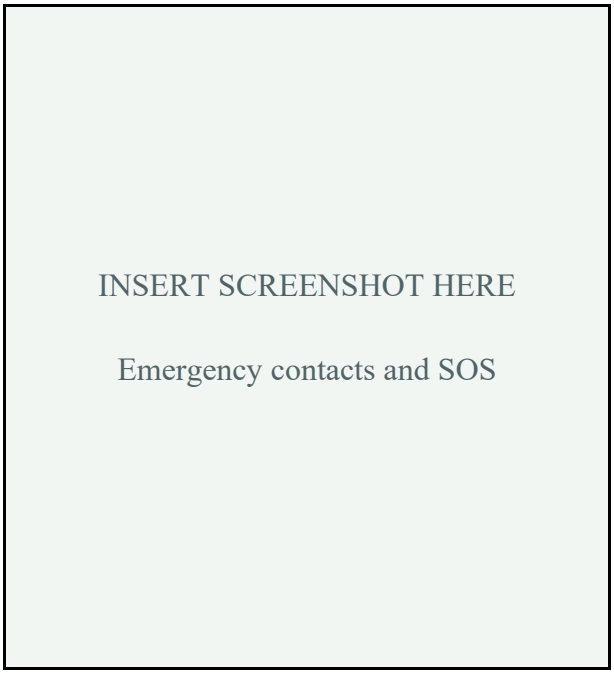


Figure 16: Emergency contacts and SOS

1. Save the caregiver, family member and doctor names and telephone numbers in Settings or the emergency-contact editor.
2. Open the emergency contact list. The order is caregiver first, family member second and doctor third.
3. Select a contact to open the supported phone dial action. Confirm the number before making the call.
4. On the watch, open SOS and tap Send SOS deliberately. Check for a successful send or a pending/error message; merely opening the screen does not send it.
5. Review the saved SOS in the phone Emergency tab, then acknowledge or respond through the alert actions. An SOS does not automatically call emergency services.

17 Elle assistant

Screen: Elle tab

What to screenshot: Capture a non-identifying question and its answer.

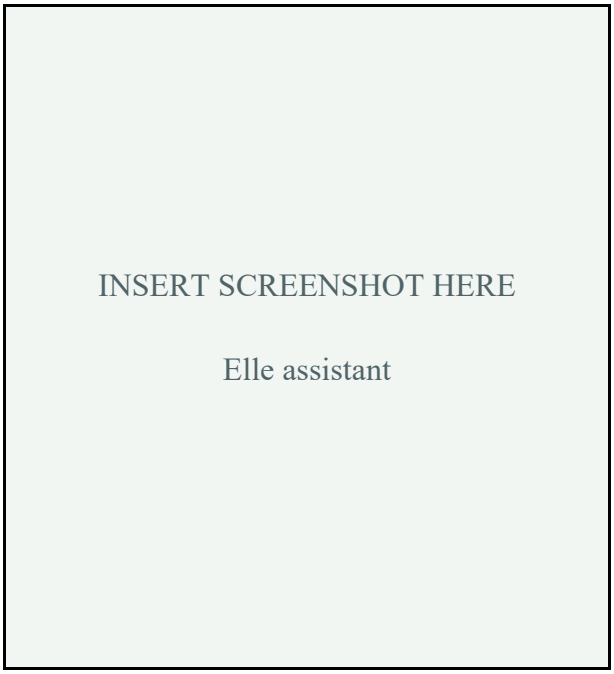


Figure 17: Elle assistant

1. Ask Elle about an existing saved record, such as a dated heart-rate reading, medication or health trend.
2. Check that the response is grounded in available patient data. Elle should not invent readings or diagnose illness.
3. For a report screenshot, use a demonstration question without private patient details.
4. Wait for the thinking indicator to finish. If a service error appears, retry while connected.
5. Use related-screen actions or evidence links when offered. Start New Chat for a separate question.

18 Settings and exports

Screen: Profile or Settings

What to screenshot: Capture watch integration, background notifications and PDF export options.

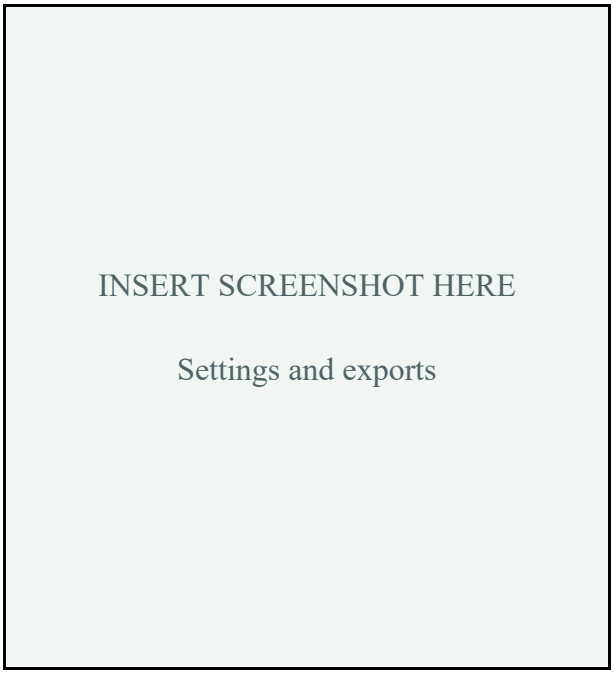


Figure 18: Settings and exports

1. Open Profile to review the active caregiver, patient and Galaxy Watch6 connection. Edit the relevant profile or emergency contacts as needed.
2. Use Health History to connect and import recorded sleep and supported overnight oxygen. Keep the patient's paired-watch identity correct.
3. Select Health vitals PDF or Medication schedule PDF. The export uses saved records for the selected patient and opens the supported sharing options.
4. If a required record is missing or an export fails, check patient selection, connectivity and the displayed message before retrying.
5. Enable background notifications from Alerts. Sign out before switching caregiver accounts and check the restored patient after the next sign-in.

19 Administrator overview and accounts

Screen: Web administrator portal

What to screenshot: Capture Overview, then Manage accounts with identifying fields covered.

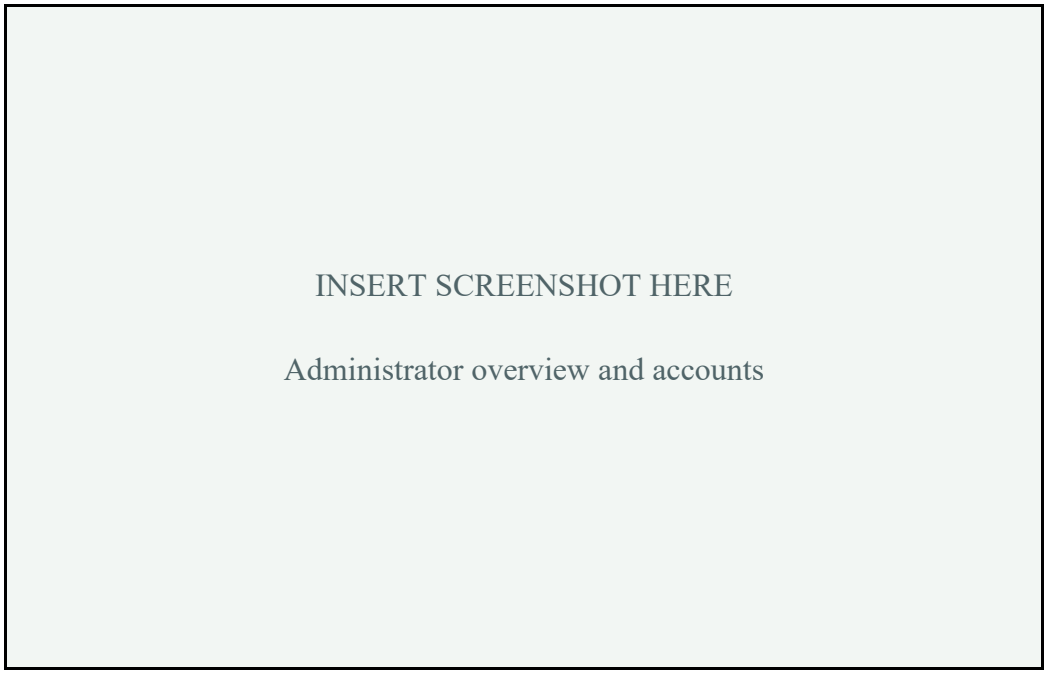


Figure 19: Administrator overview and accounts

1. Open the website's administrator portal and sign in with an authorized administrator username or email and password.
2. If sign-in fails, correct the credentials; use Forgot password and follow its verification steps when recovery is needed.
3. Overview displays saved system counts, recent alerts and emergency activity. Refresh to request updated administration data.
4. In Accounts select a caregiver and review the linked patients. Use permitted editing and access controls and read each confirmation before saving a change.
5. Use Install or try the app on the public landing page to open the Android APK and installation instructions. The caregiver phone app uses its own Google sign-in.

20 Administrator monitoring and reports

Screen: Web administrator portal

What to screenshot: Capture System monitor, Alerts and notifications, and Generate reports.

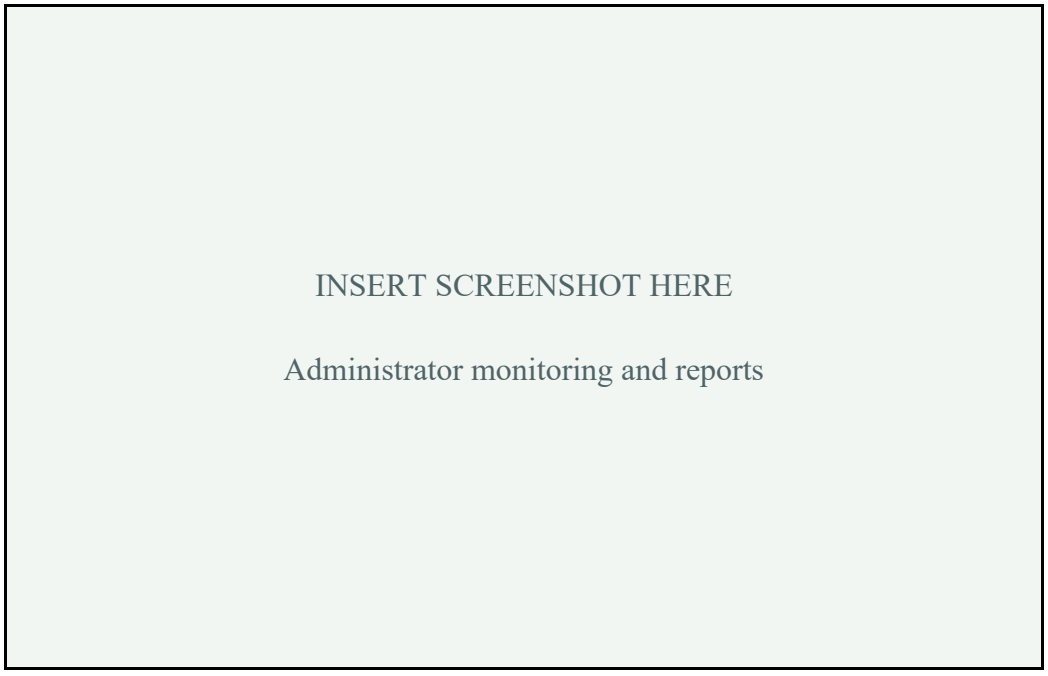


Figure 20: Administrator monitoring and reports

1. Open System monitor to inspect reported connection state and recent database health syncs. This is not a live watch sensor display.
2. Distinguish Saved at, the database write time, from Measured at, the original measurement time. Older saved readings do not establish current sensor activity.
3. Open Alerts and notifications to review saved warnings and emergency history, including status and delivery information.
4. Open Reports, choose report type and date range, then generate the available CSV. Review empty results and errors before retrying.
5. Use Security for supported administrator access and audit tasks. Sign out when finished, and hide personal identifiers in screenshots.

Installation Guide. The ElderCareAI Installation Guide provides a step-by-step process for setting up the caregiver mobile application and Galaxy Watch6 companion. It explains how to download and install the Android APK, pair the patient watch, allow monitoring permissions and configure notifications. These steps help caregivers and administrators move from installation to everyday health monitoring and care management.

1. **Download the App:** Go to https://eldercareai-capstone.vercel.app/ and select Install or try the app. Tap Download Android APK to open the supplied Google Drive file, then choose Download. If access is restricted, ask the project team to enable access to that file.

2. **Verify Requirements:** Use an Android phone running Android 10 or later for the current phone build, with available storage for the APK and app data. Keep a stable internet connection for Google sign-in, record synchronization, watch pairing and push notifications. This APK is for Android; the administrator portal runs in a web browser.

3. **Install the Mobile Application:** Open the downloaded APK through Downloads or Files. If Android requests permission, temporarily allow that browser or file manager to install unknown apps. Return to the installer, tap Install and open ElderCareAI. Turn off the temporary installation permission when finished.

4. **Update an Existing Installation:** When ElderCareAI is already installed, open the replacement APK and choose Update. Updates must use the same package and signing key. If Android reports a signature conflict, obtain a compatible signed APK from the project team before removing the existing installation.

5. **Prepare the Galaxy Watch6:** Use Galaxy Wearable to pair the patient's Galaxy Watch6 with the Android phone. ElderCareAI works best with Galaxy Watch6.

6. **Install and Set Up Samsung Health:** Install Samsung Health on the same Android phone paired to the patient's Galaxy Watch6. Open Samsung Health, complete setup and allow the requested permissions. Enable the supported sleep and blood oxygen during sleep options on the watch, and wear it while sleeping. After waking, allow the watch to synchronize with Samsung Health and check that the recorded sleep session appears there. ElderCareAI imports available records through Health History; Samsung Health does not need to stay open continuously, but synchronization and Android background restrictions can delay availability.

7. **Install the Watch Companion:** Obtain the separate ElderCareAI Wear OS APK from the project team. The phone APK is not the watch installer. For developer installation, enable watch Wi-Fi and Wireless debugging. On a computer with Android platform-tools, run adb pair WATCH_IP:PAIRING_PORT and enter the watch's displayed pairing code.

Installation Guide continued

8. **Complete Watch Installation:** Run adb connect WATCH_IP:CONNECTION_PORT, followed by adb -s WATCH_IP:CONNECTION_PORT install -r WATCH_APK_PATH. Substitute the current connection address and Wear OS APK path. The connection port is on the main Wireless debugging screen and can differ from the pairing port. Open ElderCareAI on the watch; wireless debugging may then be disabled.

9. **Sign In and Create Profiles:** Open ElderCareAI on the phone, choose Sign in with Google and select the caregiver account. Complete caregiver and older-adult profiles. Check the selected patient before connecting the watch so readings and care records are associated with the intended person.

10. **Connect the Watch to ElderCareAI:** In phone setup or Profile, select Connect or change watch and generate an eight-digit code. Enter it in ElderCareAI on the watch and tap Pair watch. Wait for the connected status. This code links the app to the patient and is separate from Bluetooth and debugging pairing.

11. **Allow Permissions and Configure Preferences:** Allow the watch's requested sensor, physical activity, precise location and notification permissions. Enable Location and keep device date and timezone correct. In the phone Alerts tab, choose Enable background notifications, grant Android permission and enable audible notification channels. Allow watch alarms and reminders when prompted for medication timing.

12. **Connect Health History:** In Profile, connect Health History and grant the requested Samsung Health access. Confirm that the same patient watch is paired to this phone. Sync after a recorded session is available. Developer builds may require the provided Samsung Health Data SDK and Sensor SDK access setup; contact the project team if authorization fails.

13. **Start Monitoring and Verify Operation:** Tap Start monitoring on the watch. Confirm fresh readings, original timestamps and upload status on the phone. Enable watch internet through Wi-Fi or supported LTE when the paired phone stays behind. Check location source and accuracy, save the safe zone, and verify a deliberate notification test. Fused location may represent the paired phone; an estimated crossing is not proof of the wearer's actual position.